

ADAM

ADAM

AUTONOMOUS DESKTOP AI MODULE

MASTER PRODUCT DOCUMENT

Product identity • brand • architecture • hardware • software • capabilities • roadmap • commercialization

Developed by Dgen Technologies Private Limited
Kolkata, India

CONFIDENTIAL — PRODUCT & PARTNER REFERENCE

Working product positioning

“A physical AI companion for your desk.”

Internal market positioning used in the launch strategy: **India's first domestically designed and built AI desk companion robot.** This is a positioning claim used by Dgen's strategy materials, not an independently verified market statistic.

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A More Human Tomorrow

ADAM — a physical AI companion designed to make everyday technology feel more natural, useful and human.

1. Product Identity

The canonical language and high-level definition of ADAM.

Field	Canonical reference
Product name	ADAM
Expansion	Autonomous Desktop AI Module
Category	Physical AI / desktop AI companion robot
Developer	Dgen Technologies Private Limited, Kolkata, India
Core interaction model	Real-time voice + vision + memory + touch + expressive face + physical movement + tool/computer control
Primary AI platform	Google Gemini Live API in the documented prototype stack
Launch model	Founder Edition → limited batches → India → international expansion

One-line definition

ADAM is a physical desktop AI companion designed to make multimodal AI tangible: it can hear, see, remember, respond, express emotion, react to touch and presence, move its head, interact with connected devices, and operate a computer through an agent layer.

Core narrative

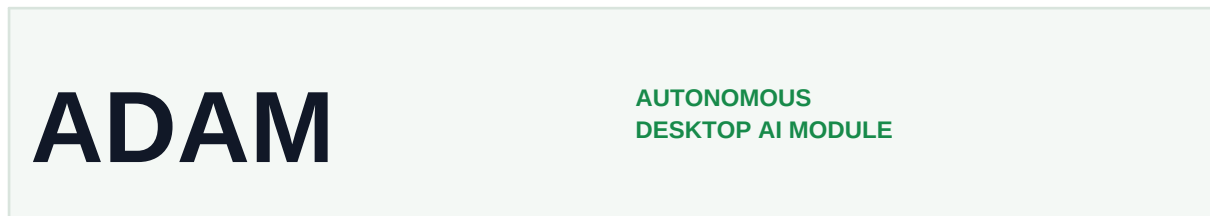
Physical AI companion + edge hardware + Gemini + vision + voice + memory + emotional interaction.

The product is intentionally positioned as more than a voice assistant in a box: the physical body is part of the interface.

2. Brand & Visual Identity

Brand references assembled from the existing Dgen/ADAM visual material.

ADAM wordmark



Visual direction

- Minimal, premium, technology-forward presentation with generous white/negative space.
- Green + dark neutral palette is consistent with Dgen's current visual identity.
- ADAM is visually associated with a compact expressive robot form, a screen-based face and subtle green accenting.
- Use the Dgen corporate identity for company-level communications; use ADAM as the product identity.

Messaging hierarchy

Layer	Use
Product	ADAM — Autonomous Desktop AI Module
Category	Physical AI desktop companion
Technology	Real-time multimodal AI, vision, voice, memory and physical interaction
Trust / platform	Powered by Google Gemini AI — where applicable to the deployment
Origin	Designed and built by Dgen Technologies in Kolkata, India

Note: the exact production logo artwork should be treated as the authoritative asset when available; this master document uses a clean text treatment so it does not invent a new logo mark.

3. System Architecture

ADAM's architecture has evolved from the working v32 prototype toward a unified custom-PCB production architecture.

CURRENT SYSTEM VIEW



Production direction: consolidate head electronics onto a unified ESP32-S3 motherboard.

Current working prototype architecture (v32)

- **Body:** Raspberry Pi Zero 2 W as the main compute/orchestration unit; audio I/O, Gemini Live session, memory, pan servo and body↔head communications.
- **Head:** ESP32-CAM for camera capture, touch/gesture sensing and tilt servo control; Raspberry Pi Pico drives the ILI9341 face display.
- **Neck link:** four wires — 5V, GND, TX and RX — over a wired UART path. The v32 engineering document explicitly confirms zero Wi-Fi for the vision link.

Production hardware direction

- **Body motherboard BOM 1:** Raspberry Pi Zero 2 W integrated with discrete power, audio ADC, amplifier, servo protection and neck interface circuitry.
- **Head motherboard:** ESP32-S3-WROOM-1-N16R8/N8R8 consolidating camera, TFT, touch, tilt servo and programming interfaces onto one PCB.
- **Advanced body option BOM 2:** Raspberry Pi Compute Module 4 Lite carrier architecture for higher-performance computer vision/OpenCV and multi-threaded AI workloads.

Important architecture distinction

The v32 document describes the **working assembled architecture**, while the September 2026 BOM documents describe the **next production PCB architecture**. They should not be presented as identical hardware revisions.

4. Feature Set

Current product feature definition — physical AI, productivity, coding and companion capabilities.

1

Real-time Conversation

Low-latency, natural voice conversation designed for continuous back-and-forth interaction.

2

Vibe Coding

ADAM can announce the status of connected coding agents such as Claude Code or Codex — including when input is required, a task is completed, or a task has stopped.

3

Smart Home Control

Control connected smart-home devices through the supported local automation layer, including lights, plugs and other MQTT-enabled devices.

4

Assistive Productivity

Help manage schedules, remember important information, add notes, draft emails and prepare text for the connected laptop or PC.

5

Companion

A persistent physical presence designed for interaction, conversation, reminders and everyday companionship.

6

Computer Assistance

Work alongside the connected laptop or PC through the computer-control and clipboard workflow.

7

Memory & Reminders

Remember user-provided information, recall it later, and assist with schedules, reminders and notes.

8

Mobile Notification Reading

When enabled by the user in the ADAM mobile app, ADAM can read incoming mobile notifications aloud.

9

Vision

Use the camera for visual context and multimodal interaction.

10

Physical Expression

Express interaction through the display, touch sensing and physical pan/tilt movement.

USER-CONTROLLED FEATURES

Mobile notification reading is opt-in. Computer, clipboard and coding-agent integrations depend on the connected device and configured permissions. ADAM reports coding-agent state rather than silently taking over the developer's workflow.

5. AI & Software Stack

Updated function-calling architecture: cloud conversation + local on-device action routing.

Runtime split

Component	ADAM implementation
Conversation AI	Gemini Live remains the primary conversational / multimodal AI layer in the documented ADAM runtime.
Function calling	Cactus Needle 3 is the planned on-device tool-calling layer for local action selection and argument filling.
Local execution	ADAM executes the selected function through its own Python/runtime integrations; Needle does not replace the executor.
Structured extraction	Needle 3 can also turn messy inputs such as notifications into typed fields for downstream actions.
Fallback	Needle 2 remains a compatibility option for existing deployments; the master architecture targets Needle 3.

ADAM function-calling domains

- Smart home — map natural language to device actions and support multi-call requests such as changing multiple devices.
- Assistive actions — schedules, reminders, notes, clipboard preparation and other connected-PC actions.
- Vibe coding — announce and route status events from Claude Code / Codex workflows, such as input required, task completed or task stopped.
- Mobile notifications — when enabled, structure notification content for reading or downstream handling.
- Memory — route save, recall and related memory operations using explicit tool schemas.
- Computer control — select narrowly defined actions for the connected laptop/PC rather than generating unrestricted commands.

Needle 3 deployment decision

Needle 3 is the current target because Cactus positions it specifically for mobile, wearables, robots, smart home and microcontroller-class devices. Its ladder allows the same model family to be built from 2 to 20 layers. For ADAM, 4-layer and 8-layer builds should be benchmarked on the Raspberry Pi Zero 2 W before the production depth is frozen.

External basis: Cactus Needle 3 release/docs, 17–18 Sep 2026; Raspberry Pi Zero 2 W official specifications.

6. Human–Robot Interaction

ADAM is designed around multimodal interaction rather than a single input channel.

Hear Dual microphone capture provides stereo audio input into the main compute/audio pipeline.	See The head camera captures frames for vision-aware AI interaction.
Feel Four touch channels provide direct physical interaction points.	Sense presence Microwave radar provides local presence awareness.
Look expressive The TFT face renders multiple emotional states and transitions.	Move Pan and tilt servos allow physical orientation and expression.
Remember Persistent memory can store user-defined information and person associations.	Act Tool calls allow ADAM to control connected software, smart-home systems and a laptop agent.

Example interaction loop

Presence detected → ADAM becomes attentive → user speaks → microphone audio enters Gemini Live → camera context may be supplied → AI responds → speaker plays response → emotion changes on the face → movement/touch/automation tools can be invoked.

7. Hardware Architecture

Consolidated from the current production BOMs and the working v32 system.

Body subsystem	Current / production reference
Compute	Raspberry Pi Zero 2 W; advanced BOM 2 option uses Raspberry Pi CM4 Lite (2GB/4GB)
Microphones	2× microphones; production BOM 1/2 documents analog condenser inputs → LMV358 → PCM1808 I2S ADC
Audio amplifier	MAX98357A 3.2W mono Class-D I2S amplifier
Speaker	Internal 3W-class acoustic speaker connection
Presence sensor	RCWL-0516 microwave radar
Pan actuator	MG90S metal gear micro servo
Power protection	USB-C input, polyfuse, reverse-polarity MOSFET, TVS, bulk capacitance and dedicated servo isolation
Neck link	4-pin JST-PH, 5V/GND/TX/RX, shielded high-flex cable

Head subsystem	Production motherboard reference
MCU	ESP32-S3-WROOM-1-N16R8 / N8R8; N16R8 is recommended for OTA partitions + LittleFS
Camera	OV2640 2MP module on 24-pin 0.5mm FPC
Display	2.4-inch 320×240 ILI9341 TFT on 20-pin FPC
Touch	4× TTP223 external modules or native capacitive touch pads
Tilt	MG90S metal gear micro servo with local bulk capacitor and flyback protection
Programming	Bottom-side pogo pads for 3V3, GND, EN, IO0, TX0, RX0, USB D-/D+

8. Production PCB Reference

Key engineering facts from the September 2026 production BOM documents.

Body Motherboard — BOM 1

- Compute core: Raspberry Pi Zero 2 W.
- 2-layer or 4-layer FR4, 1.6 mm, 1 oz copper.
- 3.0 A primary input polyfuse and reverse-polarity protection.
- 2200 μ F 16 V low-ESR input reservoir; dedicated 1000 μ F servo-local reservoir.
- PCM1808 24-bit / 96 kHz stereo I2S ADC with synchronous master-clock topology.
- MAX98357A 3.2 W mono Class-D I2S amplifier.
- 921600-baud shielded UART neck interface to the head motherboard.

Head Motherboard — ESP32-S3

- ESP32-S3-WROOM-1-N16R8 / N8R8; 240 MHz dual-core Xtensa LX7.
- 8 MB Octal PSRAM; 16 MB Flash recommended for production.
- OV2640 2 MP camera via 24-pin FPC.
- ILI9341 2.4-inch 320×240 display via high-speed DMA SPI.
- TFT control deliberately moved to GPIO37/GPIO38; GPIO45/GPIO46/GPIO3 left isolated as documented strapping-risk mitigation.
- Camera SCCB uses GPIO39/GPIO40; touch channels use GPIO1/GPIO2/GPIO4/GPIO14.
- Production head uses a unified controller rather than the prototype's ESP32-CAM + Pico split.

Advanced Body Motherboard — BOM 2

- Raspberry Pi CM4 Lite with 2 GB or 4 GB LPDDR4.
- Dual Hirose DF40 100-pin board-to-board connectors.
- 4-layer FR4 carrier with controlled-impedance routing.
- MicroSD storage, external dual-band antenna, nRPIBOOT recovery and GLOBAL_EN reset access.
- Designed for higher-performance OpenCV, computer vision and multi-threaded AI workloads.

9. Hardware Cost & Variant Reference

Engineering BOM estimates — not consumer retail pricing.

Variant	Compute	Estimated unit BOM	Role
Body BOM 1 + Head	Pi Zero 2 W + ESP32-S3	≈ ■2,830 body + ■1,538 head = ≈ ■4,368 before mechanical/assembly extras	Baseline production direction
Body BOM 2 + Head	CM4 2GB + ESP32-S3	≈ ■5,417.50 body + ■1,538 head = ≈ ■6,955.50 before mechanical/assembly extras	Advanced vision / compute
Body BOM 2 4GB + Head	CM4 4GB + ESP32-S3	≈ ■6,417.50 body + ■1,538 head = ≈ ■7,955.50 before mechanical/assembly extras	Higher-memory advanced option

Launch pricing reference from GTM strategy

- Founder drop: first 10 units, numbered Founder Editions #001–#010.
- India target price documented: ■14,999 Standard to ■18,999 Pro.
- International target price documented: \$299 Standard to \$349 Pro.
- The GTM strategy explicitly frames these as positioning targets, not final prices.

Important: BOM cost is not manufacturing cost-to-sell. Final COGS must also account for mechanical parts, PCB assembly, display/camera sourcing, battery/power system if used, enclosure finishing, packaging, QA, shipping, warranty reserve, software infrastructure and taxes.

10. AI Service & Monetisation Model

Product strategy reference derived from the Gemini billing/monetisation document.

Core principle: never brick the robot

The documented monetisation strategy treats AI usage as a service layer on top of hardware. ADAM should remain useful even if a user has no AI credits or configured API key.

Capability	Lite mode	Dependency
Clock / date / alarms	Available	Local system time
Smart-home control	Available	Local MQTT over Wi-Fi
Face detection	Available	Local Haar cascade
Presence detection	Available	RCWL-0516 hardware
Weather	Available	OpenWeatherMap API
Timers / reminders	Available	Local display/runtime
Emotion animations	Available	Local TFT renderer
Full AI conversation	Requires AI access	Gemini Live
Web search	Requires AI access	AI session/tool layer
New memory writes	AI-dependent in documented model	AI session + memory tool

Documented monetisation options

- **BYOK:** user supplies their own API key, minimizing Dgen's AI usage cost.
- **Prepaid credit packs:** users buy AI usage without being forced into a subscription.
- **Ephemeral token architecture:** documented as the production direction for multi-user Gemini Live deployment.

11. Go-To-Market & Launch Positioning

The existing GTM strategy is centered on scarcity, build-in-public storytelling and direct sales.

Phase	Objective	Key mechanism
1 — Founder Drop	Convert the first 10 strategic buyers into advocates	Numbered Founder Editions #001–#010
2 — Demand Engine	Build demand for the next batch	Build journey + Founder stories + influencer/community content
3 — Market Expansion	Expand beyond India	India first, then South Asian diaspora and global AI enthusiasts

Primary launch messages in the existing strategy

- Made in India / domestically designed and built.
- Real-time Gemini AI with physical embodiment.
- Works with Hindi and Bengali is explicitly proposed as an India-market demonstration message.
- Smart-home control: a physical AI companion that can act on connected devices.
- Direct support from the team / local product ownership.

Sales channel architecture

- Primary early channel: direct sales through dgentechologies.com.
- Secondary: Instagram/DM and direct outreach payment links.
- B2B opportunity: startups, co-working spaces and tech gifting.
- Marketplace expansion was planned only after product stability and meaningful unit volume.

12. Target Users & Product Differentiation

A factual framing of who the product is designed to serve and what its documented architecture enables.

AI enthusiasts Users who want multimodal AI to exist as a physical object rather than only an app.	Developers / makers A hackable physical-AI platform combining embedded electronics, AI, sensors and actuators.
Desk / productivity users A companion for reminders, information, voice interaction, computer actions and smart-home routines.	Smart-home users Users who want conversational control of local connected devices.
Collectors / early adopters Founder Editions create a limited early ownership experience and community identity.	

Documented comparison dimensions

- Voice-first assistants typically live in speakers or phones; ADAM adds a physical body, display, movement and touch.
- Emotion-focused companion robots may emphasize personality; ADAM's documented stack adds live AI, vision, memory, smart-home and computer-control layers.
- The product is being developed as an integrated hardware/software platform rather than a purely cloud service.

13. Product Evolution & Roadmap

A consolidated view of what has already changed and the direction reflected by current engineering files.

Stage	Architecture	Significance
Working v32	Pi Zero 2 W + ESP32-CAM + Pico + ILI9341	Confirmed physical prototype and working software path
Body BOM 1	Pi Zero 2 W on custom motherboard	Cost-optimized production-oriented integration
Head BOM	ESP32-S3 unified head motherboard	Removes legacy ESP32-CAM + Pico split
Body BOM 2	CM4 carrier	Advanced compute/vision option

Engineering priorities visible across the latest documents

- Reduce wiring and simplify the mechanical neck interface.
- Consolidate head electronics onto a production ESP32-S3 PCB.
- Improve power integrity around servos and audio.
- Use synchronous I2S clocking for reliable audio.
- Protect boot-strapping pins and isolate high-risk GPIOs.
- Prepare for OTA firmware and persistent local storage.
- Keep an upgrade path to CM4-class compute for higher-end vision workloads.

14. Technical Specification Sheet

Condensed reference updated with the planned on-device function-calling layer.

Category	Specification
Product	ADAM — Autonomous Desktop AI Module
Form	Desktop physical AI companion robot
Main compute — baseline	Raspberry Pi Zero 2 W, quad Cortex-A53 @ 1 GHz, 512 MB RAM
Main compute — advanced	Raspberry Pi CM4 Lite, 2 GB or 4 GB LPDDR4
Head MCU — production	ESP32-S3-WROOM-1-N16R8 / N8R8, 240 MHz dual-core, 8 MB Octal PSRAM
Camera	OV2640, 2 MP
Display	ILI9341, 2.4-inch, 320×240 TFT
Touch	4 channels; TTP223 or native ESP32-S3 capacitive touch
Presence	RCWL-0516 microwave radar
Actuation	MG90S metal-gear micro servos for pan/tilt
Audio input	2-microphone architecture; PCM1808 24-bit / 96 kHz stereo I2S ADC
Audio output	MAX98357A 3.2 W mono Class-D I2S amplifier
Head/body link	4-wire 5V/GND/UART; documented 921600 baud
Conversation AI	Gemini Live in the documented prototype/strategy stack
Function calling	Cactus Needle 3; 2–20 layer ladder; ADAM validation target: 4L/8L on Pi Zero 2 W
Software	Python + embedded firmware; OpenCV, memory, web search, laptop agent and display renderer
Smart home	MQTT-based local device control

15. Engineering Architecture Notes

Key implementation details, including the local function-calling boundary.

Function-calling architecture

- Gemini Live is the conversational intelligence layer; Needle 3 is a specialised local model for selecting tools and filling typed arguments.
- Needle 3 does not act as ADAM's general chat model. Its job is structured automation: request → tool selection → arguments → JSON call.
- The tool set should remain narrow and explicit: one tool per action, user-facing names, clear argument formats and constraints.
- For requests that need multiple actions, Needle 3 can emit multiple calls in order; unsupported requests can return an empty call list instead of inventing an action.
- Needle 3 responses include a confidence value. ADAM's runtime should use this to distinguish automatic execution from confirmation-required actions.
- Triggers can be used for intents that must reach a particular tool, while the final executor remains under ADAM's application control.
- Structured extraction can support mobile notification parsing and other inputs where ADAM needs typed fields rather than free-form prose.
- The initial ADAM deployment should benchmark Needle 3 at 4L and 8L on the actual Pi Zero 2 W. Cactus publishes Pi 5 measurements but does not publish a Pi Zero 2 W throughput figure in the reviewed documentation.

Audio architecture

- Production body BOMs define the Raspberry Pi/CM4 as I2S master.
- PCM1808 ADC and MAX98357A amplifier are treated as synchronized I2S slaves.
- The documented master-clock path uses GPCLK0 / GPIO4 at 12.288 MHz for a 48 kHz audio system.
- Servo power has a dedicated protected branch to reduce brownout and motor-noise risk.

Head + wiring architecture

- ESP32-S3 production design keeps GPIO45, GPIO46 and GPIO3 isolated from functional routing as documented.
- Camera SCCB uses GPIO39/GPIO40; TFT SPI uses GPIO35/GPIO36 with DC/CS on GPIO37/GPIO38 and reset on GPIO47.
- Touch channels are mapped to GPIO1/GPIO2/GPIO4/GPIO14.
- The neck interface is deliberately minimised to four electrical conductors; the moving head keeps camera/display/touch electronics local.

16. Privacy, API & Service Design Considerations

Important product-level considerations recorded in the monetisation strategy.

- Production deployments should not expose a permanent shared Gemini API key directly to every unit.
- The monetisation strategy identifies ephemeral-token architecture as the production direction for multi-user deployments.
- The strategy notes a privacy trade-off associated with certain free-tier AI usage and calls for user consent where applicable.
- User-facing settings should clearly explain AI connectivity, API-key/BYOK configuration, credit status and what functions work locally.
- The product philosophy is to preserve local functionality when cloud AI is unavailable.

This document intentionally does not define final legal/privacy policy text. Those policies should be maintained separately and reviewed before commercial release.

17. Brand, Community & Content System

The launch strategy treats the build journey itself as part of the product story.

The Build Technical progress, wiring, PCB, firmware, debugging and prototypes.	ADAM Speaks Short AI demos showing ADAM answering, remembering, using tools and expressing emotion.
The Team Founder/team stories and the human side of building ADAM in Kolkata.	Community Founder Edition announcements, waitlist milestones and community participation.
Education Explainers on Gemini Live, physical AI, smart-home integration and BYOK.	

Community architecture proposed in GTM

- ADAM by Dgen Discord community.
- WhatsApp community for the Indian waitlist.
- Early-access build updates and feature voting.
- Monthly community call / product progress session.

*The product story should consistently connect three layers: **the robot people can touch, the AI people can talk to, and the engineering journey people can follow.***

18. Source Register & Document Hierarchy

Internal product references plus the external function-calling research basis.

Internal source register

ADAM_v32_Hardware_Software_Documentation.pdf	Working prototype architecture, runtime, communications and interaction model.
ADAM_Body_Motherboard_BOM1_PiZero2W.md	Baseline production body PCB architecture, BOM, audio/power/servo design.
ADAM_Head_Motherboard_BOM_ESP32S3.md	Production head PCB architecture, camera/display/touch/servo design.
ADAM_Body_Motherboard_BOM2_Advanced_CM4.md	Advanced CM4 body option for higher-performance workloads.
ADAM_GTM_Launch_Strategy_v1.docx	Positioning, launch model, pricing targets, channels and messaging.
ADAM_Gemini_Billing_Monetisation_Strategy_v1.docx	AI billing, API architecture, BYOK/credits and Lite Mode.
ADAM_Pi_Zero_2W_Setup_Guide.md	Pi/software setup and documented runtime/emotion details.

External research basis — function calling

Cactus Needle 3 release / platform	Needle 3: 8–29 MB ladder foundation model for tool calls, structured extraction and embeddings; 2–20 layer deployment.
Cactus supported-devices guide	Linux ARM64 support; same weights can be built at different depths; Cactus advises measuring throughput on the actual device.
Cactus tool-design guide	Tool schema design, one-tool-per-action, argument grounding, triggers and confidence/routing guidance.
Raspberry Pi Zero 2 W official specifications	Quad-core 64-bit Cortex-A53 @ 1 GHz and 512 MB RAM; basis for ADAM's baseline compute constraints.

Versioning rule

Needle 3 is recorded as the current planned function-calling layer, not as a claim that production firmware has already been validated. The final layer depth and measured latency must be frozen only after testing on the target Pi Zero 2 W hardware.

19. ADAM at a Glance

A single-page summary suitable for partner conversations and internal alignment.

ADAM — Autonomous Desktop AI Module

Physical AI, made tangible.

20. Detailed Prototype Hardware & Software Stack

ADAM v32 working prototype — the proven engineering architecture before production consolidation.

20.1 Prototype hardware

Compute:	Raspberry Pi Zero 2 W is the main brain. It runs the Gemini Live session, audio I/O, memory, web search, servo pan control and the ESP32-CAM UART link.
Microphones:	2× INMP441 digital I2S microphones. The prototype captures stereo at 48 kHz through the Pi I2S interface.
Amplifier:	MAX98357A I2S Class-D amplifier. The prototype drives a ~3 W speaker.
Speaker:	Internal ~3 W speaker connected to the MAX98357A output.
Camera:	AI-Thinker ESP32-CAM with OV2640 2 MP camera. Camera frames are JPEG-compressed and sent to the Pi over UART.
Head display:	2.4-inch ILI9341 320×240 TFT driven by a Raspberry Pi Pico in the head.
Touch:	4× TTP223 touch pads connected to the ESP32-CAM.
Motion:	MG90S pan servo on the body and MG90S tilt servo in the head.
Presence:	RCWL-0516 microwave radar connected to the Pi. Presence gates attention/session behaviour.
Neck link:	4 wires across the moving neck: 5 V, GND, TX and RX. The prototype uses a wired UART vision/control link rather than Wi-Fi.

20.2 Prototype software stack

- ADAM_Pi.py is the main orchestration process and manages Gemini Live, audio, camera/gesture ingestion, servos, emotions, memory, web search and laptop control.
- Gemini Live is the conversational and multimodal AI layer. The documented prototype uses Gemini Live with Charon voice.
- Audio input: 48 kHz stereo capture → conversion to 16 kHz mono → Gemini Live.
- Audio output: Gemini's 24 kHz mono response → upsampled to 48 kHz stereo → MAX98357A → speaker.
- UART communication runs at 921600 baud between Pi and the head controller.
- Camera cadence is documented at 1 FPS in the v32 Gemini pipeline, gated by attention/speaking state.
- Local functions include time/date, emotion, save/delete/get memory, person memory, web search and laptop-agent control.
- Optional Vosk wake-word detection and idle nudges are used to control when ADAM becomes attentive.

20.3 Prototype end-to-end flow

Voice → INMP441 → Pi I2S → 16 kHz mono → Gemini Live → response audio → 48 kHz → MAX98357A → speaker.
Camera/touch → ESP32-CAM → UART → Pi. Pi emotion/control commands → UART → ESP32-CAM → Pico → TFT.
Radar → Pi → presence/session gating.

PROTOTYPE STATUS

This architecture is the current proven v32 reference. It is not the final production hardware architecture; the production design consolidates the head electronics and upgrades the audio/display strategy.

21. Detailed Production Hardware & Software Stack

Production architecture — custom body/head PCBs, consolidated head electronics and a serviceable local/cloud AI split.

21.1 Production body hardware

Baseline compute:	Custom Body Motherboard BOM 1 integrates a Raspberry Pi Zero 2 W as the cost-optimised production compute core.
Advanced compute:	BOM 2 provides a Raspberry Pi CM4 Lite option (2 GB or 4 GB) for heavier computer vision, OpenCV and multitasking workloads.
Microphones:	2× analog electret microphones → LMV358 dual preamplifier → TI PCM1808 24-bit/96 kHz stereo I2S ADC. This replaces the prototype INMP441 arrangement.
Amplifier:	MAX98357A 3.2 W mono Class-D I2S amplifier with fixed +12 dB gain configuration.
Speaker:	Internal 3 W acoustic speaker, supported in the BOM as 4 Ω or 8 Ω.
Presence:	RCWL-0516 radar remains a local, always-available presence sensor.
Power:	USB-C 5 V input, protected rails, large input reservoir and a dedicated servo protection branch to reduce motor-induced brownouts.
Body/head link:	4-core shielded high-flex cable carrying 5 V, GND, TX and RX at the documented 921600-baud UART link.

21.2 Production head hardware

- ESP32-S3-WROOM-1-N16R8 / N8R8 becomes the unified head controller, replacing the separate ESP32-CAM + Pico arrangement.
- OV2640 2 MP camera remains the baseline camera module, connected through an FPC interface.
- Production display target: a custom oval-shaped AMOLED will be developed to match ADAM's face geometry. If a custom oval panel cannot be manufactured at acceptable cost/yield, the fallback is a display with the maximum practical active area that fits the enclosure.
- The current production PCB reference supports a 2.4-inch 320×240 ILI9341 TFT as the engineering/fallback panel; the final AMOLED mechanical/electrical interface must be frozen before production PCB release.
- Four touch channels remain available through TTP223 modules or native ESP32-S3 capacitive touch pads.
- MG90S tilt servo remains local to the head.

21.3 Production software & AI boundary

- Gemini handles natural conversation, multimodal understanding and the conversational response loop when the user has Gemini access.
- Needle 3 handles local function calling: selecting the correct tool, filling typed arguments, supporting multiple calls and returning structured calls/confidence.
- ADAM's runtime remains the executor. Needle selects and structures an action; it does not independently control hardware without ADAM's permission/execution layer.
- Local memory stores persistent user-approved information and supports save/recall operations. Memory should remain useful across sessions and remain readable in the user's account/app experience.
- Presence detection is local: RCWL-0516 detects occupancy/attention conditions and can gate session startup, wake behaviour and session close without sending radar data to the cloud.
- Smart-home control uses local MQTT and therefore remains available in Lite Mode.
- Production AI access is selected during setup: BYOK, Dgen credits, or Lite Mode.

22. ADAM End-to-End Data Flow & Intelligence Routing

The key architectural rule: Gemini converses; Needle calls tools; ADAM executes; local sensors remain local.

22.1 Voice conversation flow

- User speaks → microphones capture audio → production ADC digitises the signal → Pi/CM4 audio pipeline preprocesses it.
- With Gemini enabled, audio is sent to Gemini Live. Gemini interprets the request, maintains conversational context and produces the conversational response.
- Gemini response audio returns to ADAM → I2S → MAX98357A → internal speaker.
- At the same time, ADAM updates the face/emotion state through the head controller and display.

22.2 Function-calling flow

- User request → conversational context identifies that an action is required.
- Needle 3 receives the relevant user text/context plus the available tool definitions.
- Needle selects the tool and fills its typed arguments. If the request requires multiple actions, it can produce multiple calls in order.
- ADAM evaluates the result/confidence and executes the permitted function.
- Function result returns to the ADAM runtime and, when needed, Gemini turns the result into a natural spoken response.

22.3 Presence detection & attention

- RCWL-0516 radar is connected directly to the body compute system.
- Presence is evaluated locally and can prepare/wake the assistant, maintain an active interaction window and close an AI session after the user leaves.
- Presence sensing does not require Gemini and therefore remains available in Lite Mode.
- Optional wake-word detection can provide a second gate before a conversational session begins.

22.4 Memory architecture

- ADAM maintains persistent user-approved memory for things the user explicitly asks it to remember.
- Memory operations are exposed as explicit functions such as save, delete and get memory; person-memory can be handled separately.
- Local memory remains available to the device even when Gemini is unavailable, subject to the capability limits of the selected mode.
- Memory should be surfaced through the mobile/desktop account experience so users can understand and manage what ADAM remembers.

22.5 Smart-home & assistive data paths

- Smart home: voice/request → Needle 3 → MQTT tool → local device → result → ADAM/Gemini response.
- Assistive PC action: voice/request → Needle 3 → ADAM action → paired desktop app → laptop/PC action or clipboard → result.
- Vibe coding status: desktop app observes the configured Claude Code/Codex workflow → exposes status to ADAM → ADAM can announce events such as input required, task completed or task stopped.
- Mobile notification: phone receives notification → user has enabled ADAM notification access → mobile app structures/shares the permitted notification data with ADAM over BLE → ADAM reads it aloud or processes the permitted action.

23. Mobile App & Desktop App Architecture

The apps form ADAM's user-facing control plane for setup, pairing, permissions, status and connected-device workflows.

23.1 Mobile app

- The mobile app is the primary onboarding and account-management companion for ADAM.
- During setup, the phone can pair directly with ADAM using BLE provided by the ESP32-S3 head controller.
- The Raspberry Pi's Bluetooth is intentionally disabled in the production ADAM body so the Pi's serial UART remains dedicated to the head link; BLE is handled by the ESP32 instead.
- BLE is used for local provisioning and device-to-phone data exchange: setup configuration, selected preferences, pairing information and permitted notification data can be shared with ADAM.
- The mobile app controls whether notification reading is enabled. Notification access is opt-in and can be disabled by the user.
- The app provides account state, AI access mode, credits/BYOK configuration, connected-device management and ADAM settings.

23.2 Desktop / laptop companion app

- The desktop app runs as a lightweight background utility with a system tray/menu-bar presence rather than a heavy always-open application.
- It wraps the laptop agent into an installable application and keeps the local HTTP action service and mDNS discovery available to ADAM.
- It allows ADAM to control supported computer actions such as volume, brightness and future actions, and provides the clipboard path for drafting/copying text.
- The app provides a status/dashboard surface so the user can see whether the laptop agent is running, whether ADAM is connected, recent activity/logs and pairing state.
- The app also exposes the coding-agent status layer for Claude Code and Codex, allowing ADAM to announce meaningful states such as input required, task completed and task stopped.
- The desktop app is designed for Windows/macOS/Linux, with background startup and tray controls documented for the production direction.

23.3 Desktop status model

- Connection status — ADAM paired / disconnected / reconnecting.
- Agent status — service running, paused, error or restarting.
- Coding status — Claude Code/Codex task active, waiting for input, completed or stopped.
- Action status — last requested computer action, success/failure and timestamp.
- System status — permissions, microphone/screen/accessibility state where required by the OS.
- Logs — user-readable recent events without requiring a terminal.

WHY BLE IS ON THE ESP32

The production architecture deliberately separates responsibilities: the Pi/CM4 focuses on compute, audio, networking and the wired head link; the ESP32-S3 provides the head-side BLE radio for mobile setup and local phone communication. This avoids depending on the Pi's Bluetooth stack while the Pi's Bluetooth is disabled for UART stability.

24. User Setup, AI Access Modes & Privacy Architecture

Three onboarding paths: BYOK, Dgen-managed credits, or Lite Mode.

24.1 First-time setup

- Power on ADAM → mobile app discovers/pairs with ADAM over BLE → user signs in or creates the Dgen account → device is associated with the user's account.
- User chooses an AI access mode. The choice can be changed later through the mobile/desktop settings.
- User grants only the permissions needed for the selected features, including notification access if they want ADAM to read mobile notifications and computer permissions required by the desktop agent.

24.2 Mode A — BYOK (Bring Your Own Key)

- User supplies their own Gemini API key during setup.
- ADAM uses the user's configured Gemini access for the conversational AI layer; Dgen does not need to fund that user's Gemini usage.
- Needle 3 remains the local function-calling layer for supported actions.
- BYOK is intended for technically comfortable users who prefer direct control of their AI account and key.

24.3 Mode B — Dgen Credits

- User buys prepaid AI credits from Dgen rather than managing a Gemini API key.
- Dgen's backend manages authentication/entitlement and can issue short-lived Gemini access credentials for production sessions.
- The conversational layer is Gemini; Needle 3 remains local for function calling and structured automation.
- This mode is intended to provide a managed experience plus Dgen account/service features without requiring the user to handle an API key.

24.4 Mode C — Lite Mode

- No Dgen AI service and no Gemini key/credits are required.
- Needle 3 is the local automation/intelligence layer for supported structured commands and local functions; it is not a replacement for Gemini's general conversational capability.
- Local features remain active: clock/date, alarms, timers/reminders, local smart-home MQTT control, presence detection, face/emotion animations and other offline functions.
- Previously stored memory can remain readable locally, while new cloud-dependent AI capabilities are unavailable until the user enables BYOK or Dgen credits.
- The device is never intentionally bricked because an AI service is unavailable.

24.5 Privacy and data boundaries

- Presence radar processing is local to ADAM.
- BLE provisioning and phone communication are local radio links between the mobile app and the ESP32-S3.
- Notification reading is opt-in and controlled by the mobile app.
- BYOK keeps Gemini authentication under the user's own Google/Gemini account; the user remains responsible for that account's policies and key security.
- Dgen-managed credits use the Dgen service layer for entitlement/authentication and Gemini session access; the production architecture should use short-lived credentials rather than embedding a permanent Dgen master key in each device.
- Lite Mode is designed to operate without Dgen AI calls.

20. ADAM Companion Mobile App

Integrated mobile product architecture, setup, BLE provisioning and operating surfaces.

The ADAM Companion App is the primary mobile control, setup and account surface for ADAM. It is designed as a cross-platform application using a Next.js static-export client wrapped by Capacitor, with a shared TypeScript contract layer and a Fastify API/backend path for production services.

Product architecture

- Monorepo: Turborepo + pnpm workspaces with apps/web, apps/api and apps/mobile-shell, plus shared UI, types and configuration packages.
- Web layer: Next.js 15 App Router with static export for the Capacitor shell, allowing the same application surface to support browser preview and native mobile packaging.
- Native shell: Capacitor targets Android and iOS; native plugins provide BLE, camera and secure-storage capabilities.
- Shared contracts: strict TypeScript/Zod models cover authentication, device telemetry, Wi-Fi provisioning, setup state, memory, gallery, OTA, credits and preferences.
- Backend: Fastify REST/WebSocket contract for authentication, device state, memory, gallery, OTA and credit services. The current project specification identifies the backend routes and persistence layer as implementation work still being wired.

13-stage first-run setup

- Splash → Welcome → Sign-in → BLE device discovery → Device found/pairing validation → Wi-Fi selection → Wi-Fi password → Connecting/handshake → Name device → Founder Edition reveal → AI Brain selection → BYOK/Credits/Lite configuration → Camera permission and owner face capture.
- The setup flow supports Google Sign-In or Email OTP, scans for ADAM-XXXX BLE advertisements, filters 2.4 GHz Wi-Fi networks, shows connection progress and stores setup state locally.
- Founder Edition units #001–#010 receive a dedicated recognition/reveal screen during onboarding.

BLE provisioning and connectivity

- ADAM's ESP32-S3 is the Bluetooth LE endpoint used for mobile provisioning and nearby mobile communication.
- The Raspberry Pi's Bluetooth is intentionally turned off in the ADAM architecture; BLE responsibilities remain on the ESP32 side rather than depending on the Pi Bluetooth controller.
- The mobile app discovers an ADAM-XXXX advertisement, validates the device/serial, then uses BLE to transfer provisioning data and establish the initial setup relationship.
- After Wi-Fi provisioning, local-network discovery can use mDNS/Zeroconf to locate adam.local. A future production bridge can maintain a WebSocket for live telemetry.

Connectivity principle

Mobile → BLE → ESP32-S3 for provisioning/nearby control. ADAM's Pi → Wi-Fi/Ethernet-network services for normal cloud/local operation. Pi Bluetooth remains disabled.

21. Mobile App — Main Product Surfaces

Post-setup controls, memory, gallery, settings and device management.

Home / live status

- Live online/offline status for ADAM, with the intended future real-time telemetry bridge for emotion state, audio activity and system vitals.
- Battery/status indicators, current emotion/face state and sleep/wake quick controls.
- The app is intended to become the user's at-a-glance operational view rather than a duplicate of the desktop application.

Memory

- Searchable knowledge stream containing stored user facts and memory entries.
- User-facing edit/delete controls for stored memories, supporting transparency and correction.
- Memory data types are represented in the shared TypeScript contracts alongside conversation logs.
- Memory can therefore be managed from the app while ADAM's local AI/runtime uses the resulting memory context during interaction.

Gallery

- Photo grid of captured moments with date grouping and detail views.
- Gallery metadata and captured media are represented in shared contracts; cloud backup is planned through the backend API.
- The production design can keep camera media local by default while making backup an explicit service/setting.

Smart home

- Room-based device management for lights, plugs, fans and other MQTT-connected devices.
- Scene shortcuts such as Movie Night and Wind Down.
- Routine automation editor, including presence-based triggers using ADAM's RCWL-0516 sensor.
- The smart-home UI is defined in the product specification but is currently a remaining implementation screen rather than a completed production bridge.

Settings & device management

- Software updates: OTA check, changelog and staged update progress.
- Laptop management: paired companion laptop/developer machine list.
- Wi-Fi: current network details and future re-provisioning without a factory reset.
- Voice: voice selection, language hint, wake sensitivity and personality controls.
- Account: profile, sign-out, Google unlink, data export and device ownership transfer.
- AI Brain: switch between BYOK, Dgen Managed Credits and Lite Mode after onboarding.

22. AI Access, Desktop App & Cross-Device Experience

How mobile setup, Dgen services, Needle 3 and the desktop agent fit together.

Three AI setup choices

BYOK — Bring Your Own Key

User supplies their own Gemini API key. The app guides setup, supports clipboard/deep-link assistance and stores the credential using native secure storage when implemented. AI usage is billed directly to the user's chosen provider account.

Dgen Managed Credits

User purchases Dgen-managed AI credits. This path is intended to provide a simpler setup plus Dgen-managed privacy/service features, usage accounting and product integrations.

Lite Mode

No Dgen-managed AI service is required. Needle 3 handles supported local function-calling/automation, while offline/local features remain available. Full cloud conversation and other cloud-dependent capabilities are unavailable without an AI provider.

AI responsibility split

- Gemini handles the natural conversation and multimodal conversational intelligence.
- Needle 3 handles local function calling: selecting the appropriate tool, filling typed arguments, supporting ordered multi-call requests and exposing confidence for routing.
- ADAM's application runtime remains the executor. Needle 3 does not receive unrestricted control of the robot; the runtime decides which permitted tools can actually execute.
- Mobile configuration determines which AI access mode is active; the device/runtime applies the corresponding service policy.

Desktop companion app

- A background/tray application connects the user's laptop/PC to ADAM and provides the computer-control bridge.
- It exposes connection/device status, agent status and operational events rather than requiring the user to keep a full window open.
- Vibe-coding integration can surface Claude Code or Codex status events to ADAM, including input required, task completed, task stopped and similar states.
- The desktop bridge supports assistive workflows such as preparing text and sending it to the connected computer's clipboard, helping draft mail and performing approved computer actions.
- Paired laptops are represented in the mobile app, allowing users to see and manage which machines are connected to ADAM.
- The desktop architecture is intended to use local discovery such as mDNS and a persistent local/WebSocket bridge for real-time state.

Cross-device model

Mobile app = setup, account, AI access, memory, gallery and device management. Desktop app = computer/agent bridge and coding status. ADAM = physical interaction, conversation, sensors and local execution. BLE is anchored on the ESP32-S3, while the Pi remains the main AI/runtime computer.

23. Mobile App Implementation Status & Roadmap

What is specified/implemented versus what remains for the production companion app.

Specified / implemented in the current project

- Turborepo/pnpm monorepo structure, shared UI/types/config packages and strict TypeScript validation.
- All 13 first-run setup screens are defined/implemented in the web client, including BLE discovery UI, Wi-Fi provisioning UI, AI Brain selection, BYOK, credits and camera/face-capture flows.
- Home, Gallery, Memory, Settings, Software Update, Laptops and About surfaces are defined.
- Mock fixtures exist for BLE discovery, Wi-Fi handoff, memory logs and OTA lifecycle testing.
- Next.js static export is configured for the Capacitor shell.

Remaining engineering work

- Replace browser BLE mocks with the native Capacitor BLE plugin and connect to the actual ESP32-S3 provisioning service.
- Implement native secure storage for API keys, tokens and credentials.
- Implement mDNS/local-network discovery and the real ADAM telemetry WebSocket.
- Complete the Smart Home, AI Brain settings, Wi-Fi, Voice and Account screens.
- Build the Fastify backend, persistence models, credit ledger/Razorpay integration and production client API wiring.
- Implement real-time emotion/audio/system telemetry and low-latency media preview.
- Add setup error handling for BLE disconnects, wrong Wi-Fi credentials and Bluetooth permissions, plus automated E2E testing.

Mobile App product principle

The app should configure and extend ADAM — not become a requirement for every interaction.

Once setup is complete, core physical interaction remains on ADAM itself. Mobile is the control plane for provisioning, preferences, memories, media, AI access, device management and selected notifications. Lite Mode is specifically intended to preserve useful local operation when Dgen's cloud service is unavailable.